

Practice Month Math Set 006

Question N1

If $6w + 1 \leq 13$, what is the greatest possible value of $6w - 1$?

- A) 1
- B) 2
- C) 11
- D) 12

Question N2

if $2x - 1 = 3a$, what is the value of $4x + 4$?

- A) $6a + 2$
- B) $6a + 3$
- C) $6a + 5$
- D) $6a + 6$

Question N3

A line has a slope of $\frac{3}{5}$ and passes through the point $(10,5)$. Which equation represents this line?

- A) $y = \frac{3}{5}x - 1$
- B) $y = \frac{3}{5}x + 5$
- C) $y = \frac{3}{5}x + 7$
- D) $y = \frac{3}{5}x + 11$

Question N4

In the xy -plane, line p is represented by the equation $y = \frac{3}{2}x - 6$. If line q perpendicular to line p and passes through the point $(-4, 1)$, which of the following is the equation of line q ?

- A) $y = -\frac{2}{3}x - \frac{5}{3}$
- B) $y = -\frac{2}{3}x + \frac{11}{3}$
- C) $y = -\frac{3}{2}x - 5$
- D) $y = -\frac{3}{2}x + 7$

Question N5

In the xy -plane, line k passes through the points $(-a, b)$ and $(b, -a)$, where $a + b \neq 0$. What is the slope of line k ?

- A) -1
- B) 0
- C) 1
- D) The slope is undefined.

Question N6

$$5x + y = 17$$

$$4x - y = 10$$

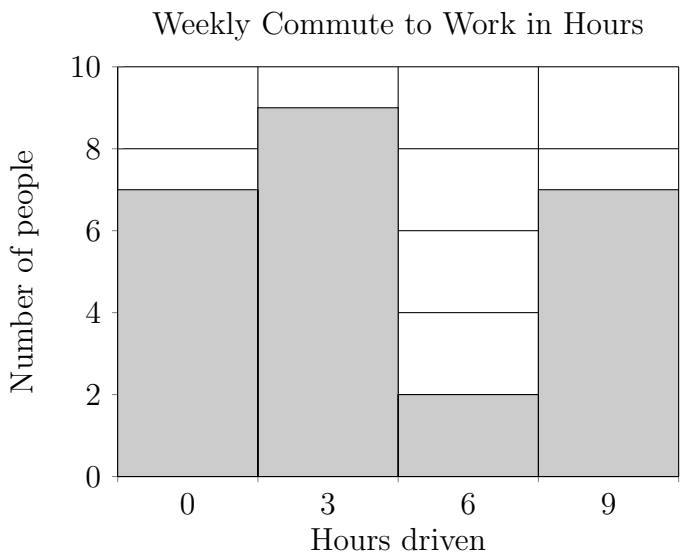
If (x, y) is the solution to the system of equations above, what is the value of $x + 2y$?

Question N7

A quality control manager at a food manufacturer is tasting the weight of potato chip bags produced by the company. The manager selects 200 bags at random from the final output that day and determines the average weight has a 95% confidence interval of 156 to 161 grams. Which of the following conclusions is most valid based on the confidence interval?

- A) The true average weight of all potato chip bags produced by the company that day is probably between 156 and 161 grams.
- B) The true average weight of all potato chip bags ever produced by the company is probably between 156 and 161 grams.
- C) 95% of all potato chip bags produced by the company that day have a weight between 156 and 161 grams.
- D) 95% of all potato chip bags ever produced by the company have a weight between 156 and 161 grams.

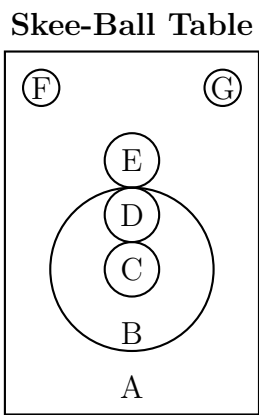
Question N8



A survey asked a group of people to estimate the number of hours they commute to work per week. The results of the survey are shown in the histogram above. Which interval contains the median number of hours driven?

- A) 3 hours
- B) 6 hours
- C) 9 hours
- D) 12 hours

Question N9



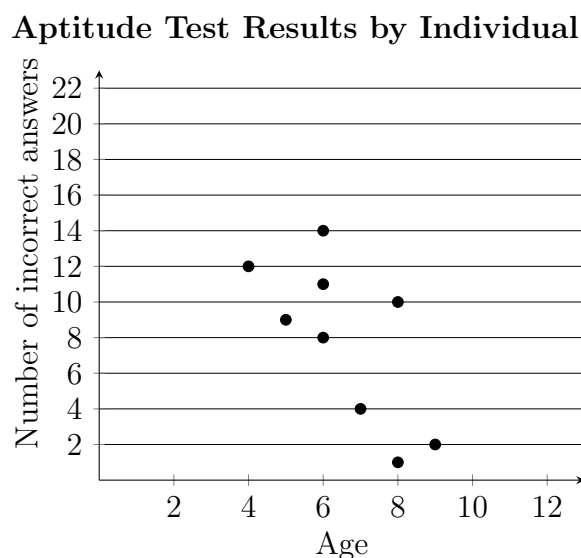
Section	Point value	Radius (in)
A	10	–
B	20	9
C	30	3
D	40	3
E	50	3
F	100	2
G	100	2

In the game of skee-ball, players try to score the highest point value by rolling a ball down a lane that then angles upward and launches the ball into one of seven sections; the section are represented in the skee-ball diagram shown above. For each turn, the player

earns the number of points designated for the section in which the ball comes to a rest. The rectangular skee-ball table has a length of 38 and a width of 28 inches, and is divided into 7 sections, A through G. Sections B through G are bound by circles, and section A is the remaining portion of the table. The radius and point value of each section are shown in the table above, (Note: Because section A is not a circle, it does not have a radius.) If Simon rolled the ball such that it landed on the board at random and he scored at least 20 points, which of the following is closest to the probability that the ball landed in section B?

- A) 0.827
- B) 0.643
- C) 0.240
- D) 0.186

Question N10



An aptitude test was given to a group of children ranging in age from 4 to 9. The scatterplot above compares the child's age with the number of incorrect answers for each child. Which of the following equations best models the data shown in the scatterplot?

- A) $y = 2x + 21$
- B) $y = 2x + 10$
- C) $y = -2x + 21$
- D) $y = -2x + 10$

Question N11

If $x^2 + y^2 = w$ and $xy = z$, which of these expressions is equivalent to $9w + 18z$?

- A) $(9x + 18y)^2$
- B) $(9x + 9y)^2$
- C) $(3x + 3y)^2$
- D) $(3x + y)^2$

Question N12

$$30x^4 - 10x^3 + 15x^2 - 5x$$

Which of the following expressions is equivalent to the expression above?

- A) $(3x - 1)(2x^2 + 1)$
- B) $x(5x + 1)(6x^2 - 5)$
- C) $5x(3x - 1)(2x^2 + 1)$
- D) $5x(3x + 1)(2x^2 - 1)$

Question N13

x	$f(x)$	$g(x)$
1	3	1
2	5	6
3	7	4
4	9	9
5	11	7
6	13	12
7	15	10

Various values of the functions $f(x)$ and $g(x)$ are given in the table above. What is the value of $g(f(2))$?

Question N14

$$f(x) = 4x^2 + px + p$$

In the polynomial $f(x)$ above, p is a constant. If $x + 3$ is a factor of the polynomial, what is the value of p ?

Question N15

What is the sum of the solutions to $(3x + 8)(2x - 5) = 0$?

- A) $-\frac{3}{5}$
- B) $-\frac{1}{6}$
- C) $\frac{1}{6}$
- D) $\frac{3}{5}$

Question N16

$$m(x^2 + 1) = 5 + m$$

In the equation above, m is a nonzero constant. Which of the following inequalities represents all values of m for which the equation has no real solutions?

- A) $m > 5$
- B) $m > 0$
- C) $m < 0$
- D) $m < 5$

Question N17

$$y = x^2 - 6x + 11$$

$$y = -5x + 13$$

If (x, y) is a solution to the system of equations above, what is one possible value of $y - x$?

Question N18

$$\sqrt{x + 5} = 1 - x$$

What is the solution set of the equation above?

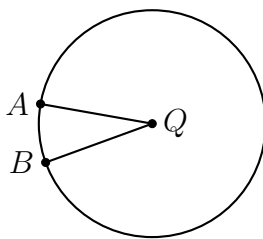
- A) $\{-1\}$
- B) $\{4\}$
- C) $\{-1, 4\}$
- D) No solution

Question N19

$$(x + 4)^2 + (y - 3)^2 = 4$$

The equation above represents a circle in the xy -plane. If the circle is shifted 3 units to the left and 3 units down and the length of its radius is halved, which of the following is an equation of the transformed circle?

- A) $(x + 1)^2 + (y - 6)^2 = 1$
- B) $(x + 1)^2 + (y - 6)^2 = 2$
- C) $(x + 7)^2 + y^2 = 1$
- D) $(x + 7)^2 + y^2 = 2$

Question N20

Note: Figure not drawn to scale.

In the figure above, Q is the center of the circle. If the measure of acute angle AQB is between $\frac{\pi}{7}$ radians and $\frac{3\pi}{19}$ radians, what is one possible integer measure, in degrees, of acute angle AQB ?

Answer Sheet

Module 1

1	2	3	4	5	6	7	8	9	10
<i>C</i>	<i>D</i>	<i>A</i>	<i>A</i>	<i>A</i>	7	<i>A</i>	<i>A</i>	<i>B</i>	<i>C</i>

11	12	13	14	15	16	17	18	19	20
<i>C</i>	<i>C</i>	7	18	<i>B</i>	<i>C</i>	1, 19	<i>A</i>	<i>C</i>	26, 27, 28